microcontrollers, memory can become the most significant limitation. In this case, the device needs enough flash memory to store two firmware images when using the A/B partitioning technique. For memory-limited devices, some developers prefer a single-partition update technique in which the current firmware is overwritten. This technique reduces memory consumption but increases the risk of an update failure.

Securing and validating OTA firmware updates

Security is a vital component of any OTA firmware update mechanism. Before installation, the device must confirm the authenticity of the package and ensure that it has not been altered during transmission. Modern OTA systems can follow a zero-trust approach in which each stage of the update process is authenticated and verified.

This starts with authenticating the device and the update server. The firmware is transmitted over an encrypted communication channel, usually using transport layer security. The firmware image is also digitally signed, and the authenticity of the signature is verified by the device before installation. Combined with secure boot, this helps ensure that only authenticated firmware is installed on the device. Firmware that fails verification is rejected.

Cybersecurity legislation has imposed stringent requirements for authentication and encryption in OTA processes. A secure OTA mechanism should authenticate the requesting device and validate the identity of the update server. Communication with the server should always use an encrypted connection to protect firmware from unauthorised access or alteration.

Security alone, however, is not sufficient. Every firmware release must also be validated before deployment to production devices.

Should companies build their own OTA platform or use a commercial solution?

- A product lifecycle management (PLM) platform can provide a more suitable foundation for managing OTA updates.
- OTA systems need detailed device information, including hardware version, product variant, production batch, and firmware history.
- This data should remain available as products move from engineering to manufacturing and then to operations.
- Since OTA updates are mainly handled during the operational stage, access to complete product lifecycle information is important for accurate device targeting.
- A PLM-based approach helps ensure that the appropriate firmware reaches the intended devices, reducing update-related risks.

Instead of making new firmware immediately available, organisations typically introduce a validation stage where each release is tested across multiple firmware versions and update scenarios to confirm successful installation, compatibility, and stability.

In a typical deployment workflow, new firmware is first uploaded to an internal software platform but remains unavailable for OTA deployment until it is approved by the support and validation team. The team performs manual testing across different firmware versions and update iterations to verify that the release functions as expected. Only after successful validation is the firmware marked as OTA-enabled, so that only tested software is deployed to field devices.

Metadata enables targeted and reliable firmware deployment

Metadata helps ensure that every device receives the right firmware update. In a mass deployment, devices may be produced for different customers, regions, use cases, or production batches. Even though most devices have similar hardware, they may need different firmware versions depending on their usage.

To address these differences, manufacturers store metadata for their devices in the product lifecycle management system. The metadata includes the manufacturing batch, hardware revision, firmware version, customer, region, and use-case profile. All this together creates a digital profile for each device.

Once the device connects to the update service platform, it compares its metadata to the records on the server. The update service provides firmware to the device only if the update corresponds to its hardware, configuration, and deployment requirements. For instance, if an update is intended for a particular customer or production batch, only devices matching those criteria can receive it.

The PLM system allows manufacturers to target updates at different levels of detail, from an entire product family to a specific hardware revision, production batch, or even individual device.

Before installation, the device uses its metadata to verify that the firmware is compatible with its hardware and configuration. If the validation fails, the update is rejected, preventing incompatible firmware from being installed.

Minimising downtime and scaling firmware updates

One of the key goals of a FOTA system is to update devices without disrupting their normal operation. This is especially important in industrial environments, where connected devices are deployed across factories and production lines. Updating a device while it is actively